

Agent-Agnostic Guardrails: Universal Java Code Quality with AGENTS.md and Static Analysis

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Agenda

1.  The Problem
2.  Context Engineering
3.  AGENTS.md and how to draft it
4.  Checkstyle · SpotBugs · ArchUnit
5.  Agent Feedback Loop using static analysis
6.  Live Demo
7.  Call to Action

The Problem: Agent Inconsistency

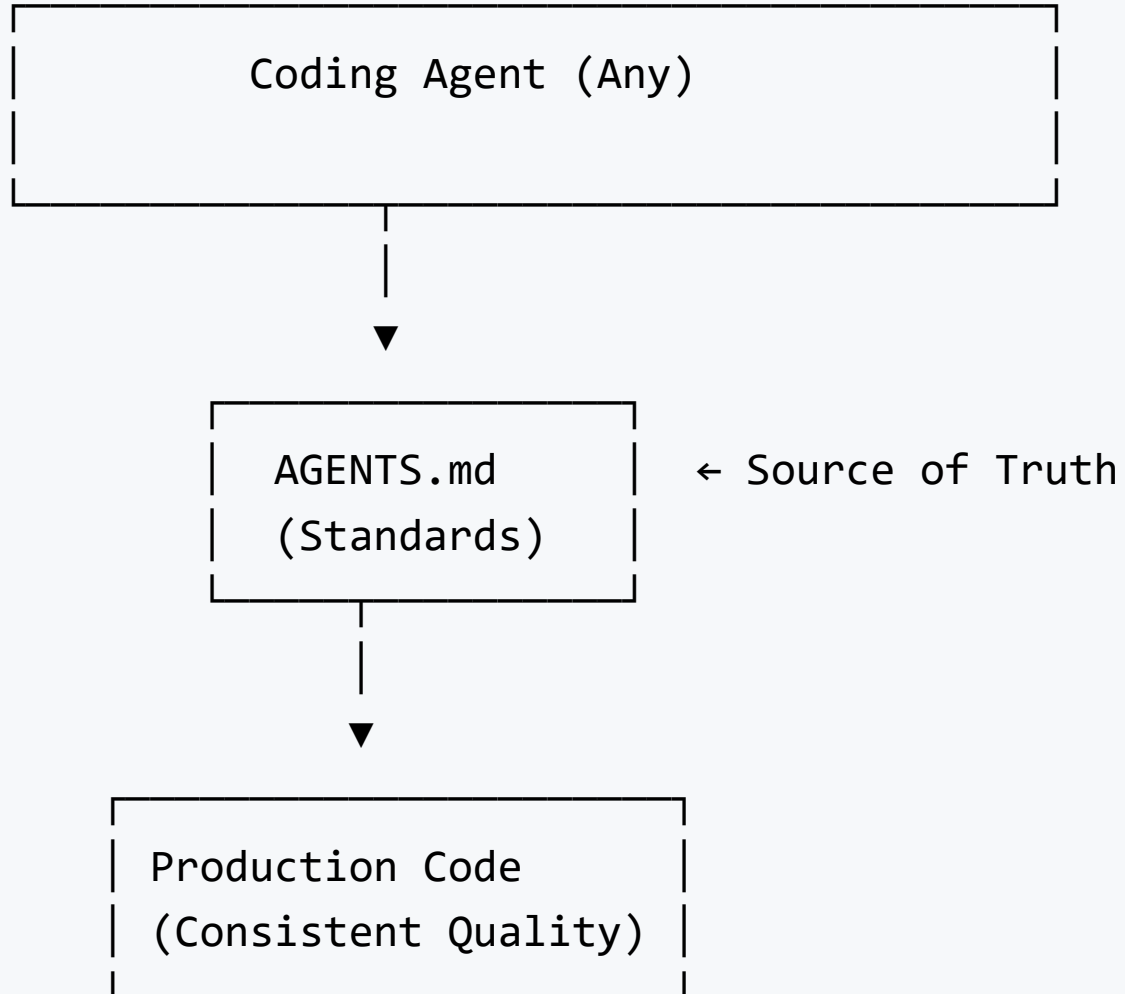
AI coding agents are powerful, but unpredictable.

Challenges:

-  Different agents write code differently
-  Each agent has different rule definitions
-  No consistency across agent versions
-  No accountability for code quality
-  Prompt changes = different outputs

Result: Production code that's hard to review, maintain, and validate.

The Solution: Agent-Agnostic Standards



Context Engineering for Coding Agents

| "Curate what the AI sees to reduce guessing."

- Brief it like a new teammate
- Provide rules + constraints + examples
- Better context → fewer surprises

Key Insight: `AGENTS.md` is one piece of the puzzle.

Context Window / What AI sees

System Instructions & Behavior Guidelines

Agent rules, safety guardrails, base behavior — set by the tool or vendor



Custom Instructions

Project-specific rules — your AGENTS.md, coding standards, team conventions



Conversation History

Everything in the current session — prompts, replies, corrections



Implicit Context

Open files, selected code, git diff — silently passed by your IDE



Explicit References

Files or docs you manually attach — #file, #codebase, pasted snippets



Tool Outputs

Terminal results, search outputs, API responses — live feedback fed back in

What You Control

- Prompt: task + constraints
- Standards: `AGENTS.md` + style rules
- Context: architecture + examples
- Tools: build/test/lint feedback

What you don't control: model reasoning or perfectly repeatable output.

What is AGENTS.md?

A human-readable, machine-executable format for defining coding standards.

Key Characteristics:

-  Open, markdown-based format
-  Works with any AI coding agent
-  Readable by humans AND AI
-  Enforceable by build systems

Learn more: <https://agents.md/>

AGENTS.md Structure

Project Overview

Build and Test Commands

Code Quality and Style

- Public methods must have JavaDoc comments
- No null pointer dereferences
- Follow resource-leak prevention patterns

Architecture

- Dependency injection required
- No static initialization blocks
- Immutable objects for data transfer

Security Consideration

- Ensure logs do not contain sensitive data / PII
- Validate all user inputs to prevent injection attacks (e.g., SQL injection, XSS)

How Agents Discover AGENTS.md

Agent starts here

|

v

```
project-root/          <- 1. Read first (global rules)
+-- AGENTS.md  [ROOT]
+-- src/
    +-- main/          <- 2. Read next (module rules)
    |   +-- AGENTS.md  [MODULE]
    |   +-- java/
    |       +-- com/myapp/service/    <- 3. Read last (nearest wins)
    |           +-- AGENTS.md  [LOCAL] <-- most specific, overrides parent
    |           +-- UserService.java <-- file being generated
```

Rules merge top-down — nearest **AGENTS.md** overrides parent on conflict

Why Static Analysis Tools?

| AGENTS.md defines standards. Static tools enforce them.

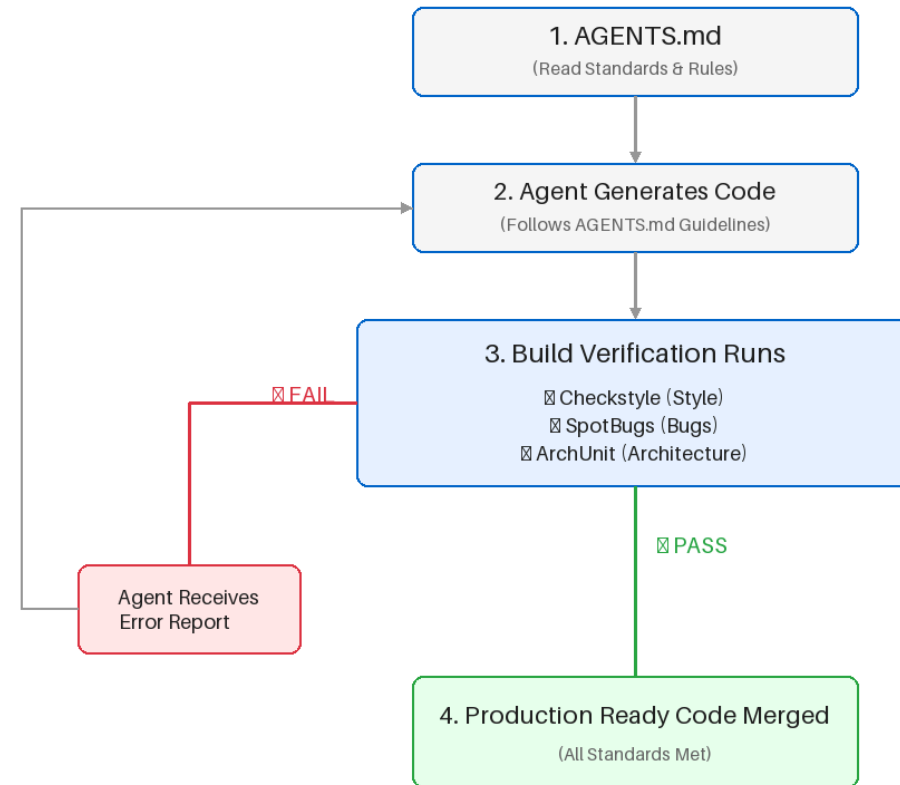
The Gap:

-  Standards exist only in documentation
-  Agents might miss subtle requirements
-  Manual review is slow and subjective
-  Violations slip into commits

The Solution:

-  Automated, measurable quality metrics
-  Manual review can focus on functional logic
-  Prevents non-compliant code from merging

Agile Agent Workflow



Problem: Code Style Inconsistency

Without Standards:

- ✗ Inconsistent naming conventions
- ✗ Varying indentation and formatting
- ✗ Random brace placement
- ✗ Difficult code reviews

Impact: Maintenance nightmare, inconsistent codebase, Low Readability

Solution: Checkstyle

What it does:

-  Enforces code style & formatting rules
-  Validates naming conventions
-  Checks indentation, whitespace, line length
-  Runs in CI/CD pipeline

Problem: Runtime & Security Bugs

Without Standards:

- ✗ Null pointer exceptions
- ✗ Resource leaks (unclosed streams)
- ✗ Type casting errors
- ✗ Integer overflow issues
- ✗ Concurrency problems

Impact: Production outages, security breaches

Solution: SpotBugs

What it does:

-  Detects potential null pointer dereferences
-  Finds resource leaks (files, connections)
-  Identifies type mismatches
-  Catches integer overflow/underflow
-  Reports concurrency bugs

Problem: Architectural Violations

Without Standards:

- ✗ Circular dependencies
- ✗ Layering violations
- ✗ Unwanted module interactions
- ✗ Business logic in wrong places
- ✗ Coupling between unrelated components

Impact: Unmaintainable systems, difficult refactoring

Solution: ArchUnit

What it does:

-  Enforces architectural rules as tests
-  Validates layer separation
-  Prevents circular dependencies
-  Controls access between modules
-  Documents architecture in code

AGENTS.md with Plugins Section

Project Standards

Code Quality Rules

- All public methods must have JavaDoc
- Maximum method length: 50 lines
- Cyclomatic complexity ≤ 10
- No null pointer dereferences
- Resources must be auto-closed

Architectural Rules

- Controller $\rightarrow$ Service $\rightarrow$ Repository (layered)
- No circular dependencies
- Service layer exceptions must extend CustomException

Plugins

Mandatory validation tools integrated with this AGENTS.md:

Tool	Config	Purpose
-----	-----	-----
Checkstyle	checkstyle.xml	Code style enforcement
SpotBugs	spotbugs.xml	Runtime bug detection
ArchUnit	ArchitectureTest.java	Architecture validation

Integration Instructions

1. Agent reads AGENTS.md as standard contract
2. Agent generates code accordingly
3. Build runs: ``./mvnw verify``
4. If violations found, agent receives error report
5. Agent regenerates fixes and retries

Agent Feedback Loop

When Build Fails:

Build Error Report:

```
|— Checkstyle: Line 45 - Missing JavaDoc  
|— SpotBugs: Potential null pointer at line 78  
|— ArchUnit: Controller accessing repository directly
```

Agent Actions:

1.  Reads error report
2.  Reviews AGENTS.md rules
3.  Regenerates code with fixes
4.  Commit passes next build

Key Benefits

✓ Agent Independence

- Any coding agent produces consistent code
- Agents become replaceable

✓ Build Accountability

- No code ships without passing standards
- Metrics are objective, not subjective

✓ Team Confidence

- Code reviews become predictable
- Maintenance is easier

✓ Enterprise Ready

Why This Matters

Prompts are fragile. Standards are forever.

-  Agents evolve quickly
-  Teams grow
-  Requirements change
-  Production code must be stable

Solution: Define standards once, use with any agent.

Live Demo Roadmap

1. 🤖 Generate code using prompt for a crud API (attempt 1)
2. 🔍 Observe the code with Checkstyle, SpotBugs, ArchUnit
3. ✍️ Now Write AGENTS.md with standards
4. 🤖 Generate code with Agent (attempt 2)
5. ❌ Build fails - violations detected
6. 🛠️ Agent receives feedback
7. ✅ Agent regenerates - build passes
8. 📊 Compare code quality metrics

Summary

Component	Purpose
AGENTS.md	Human & machine-readable standard contract
Checkstyle	Enforces code style consistency
SpotBugs	Prevents runtime/security bugs
ArchUnit	Maintains architectural integrity

Integration: Standards + Tools + Feedback Loop = Production-Ready AI Code

Call to Action

Not: "How do I prompt better?"

But: "How do I enforce standards?"

-  Standard-driven development
-  Measurable quality
-  Team scalability
-  Enterprise adoption

About the Author

Download the sample AGENTS.md and recreate the project using prompts shared in README.md!

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 GitHub: <https://github.com/ShuchitaTechTalks/agent-agnostic-guardrails>

Get the code, recreate the project, and join the conversation on building production-ready AI-generated code.

Thank You

Questions?

Resources:

- AGENTS.md: <https://agents.md/>
- Checkstyle: <https://checkstyle.sourceforge.io/>
- SpotBugs: <https://spotbugs.readthedocs.io/>
- ArchUnit: <https://www.archunit.org/>

Let's make AI-generated code boring, predictable, and production-ready.